

Milestone-3 Report: Model Optimization, Visualization & Evaluation

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1. Project Description

The **CliniScan – Chest X-ray Analysis System** is a deep learning-based medical imaging solution designed to detect and classify abnormalities in chest X-ray images.

The system consists of two main components:

- **Classification Model (EfficientNet-B2)** → predicts disease labels
- **Detection Model (YOLOv8)** → localizes abnormalities using bounding boxes

The goal of Milestone-3 is to improve model performance using:

- Hyperparameter tuning
- Data augmentation
- Transfer learning
- Model evaluation and interpretability

2. Dataset Used

The project uses the **VinBigData Chest X-ray dataset**, which contains:

- High-resolution chest X-ray images
- Bounding box annotations for abnormalities

- Multiple disease categories

Dataset structure:

```
yolo_dataset/  
├── images/  
│   ├── train/  
│   └── val/  
└── labels/  
    ├── train/  
    └── val/
```

The dataset contains **11 abnormality classes** for detection.

3. Environment Setup

The project was implemented using **Python in VS Code / Kaggle environment**.

Libraries Used:

- PyTorch, Torchvision → deep learning models
- Ultralytics YOLOv8 → object detection
- Albumentations → data augmentation
- OpenCV, PIL → image processing
- NumPy, Pandas → data handling
- Matplotlib → visualization

GPU acceleration was used to speed up training.

4. Data Exploration

Initial dataset exploration included:

- Verifying dataset structure
- Checking image-label alignment
- Visualizing sample X-ray images
- Inspecting bounding box annotations

This ensured correct dataset preparation before training.

5. Data Preprocessing

The preprocessing pipeline included:

5.1 Image Loading

Images were loaded using OpenCV/PIL and converted to RGB format.

5.2 Resizing

Images were resized to **640×640** for YOLO training.

5.3 Label Formatting

Bounding box annotations were converted into YOLO format:

class x_center y_center width height

5.4 Normalization

Pixel values were normalized for stable training.

6. Proposed System

The system architecture consists of two modules:

Detection Pipeline:

1. Input X-ray Image
2. Preprocessing (resize, normalize)
3. YOLOv8 Model
4. Bounding Box Prediction

Classification Pipeline:

1. Input Image
2. EfficientNet-B2
3. Disease Prediction

7. Model Training & Optimization

7.1 Baseline Model (YOLOv8s)

- Image size: 640
- Epochs: 30

Results:

Metric	Value
Precision	0.0031
Recall	0.181
mAP50	0.0089

7.2 Hyperparameter Tuning

Experiments performed:

- Optimizer: AdamW
- Batch size tuning
- Increased epochs

7.3 Data Augmentation

Albumentations techniques:

- Horizontal flip
- Rotation ($\pm 10^\circ$)
- Brightness/contrast
- CLAHE
- Blur

Results:

Metric	Value
Precision	0.00456
Recall	0.203
mAP50	0.0071

7.4 Transfer Learning (YOLOv8m)

- Backbone upgraded to YOLOv8m
- Training: 40 epochs

Results:

Metric	Value
Precision	0.00556
Recall	0.161
mAP50	0.00483

8. Evaluation Metrics

Detection Metrics:

- Precision
- Recall
- mAP@50
- mAP@50–95

Comparison Table:

Experiment	Precision	Recall	mAP50
Baseline	0.0031	0.181	0.0089
Augmentation	0.00456	0.203	0.0071
Transfer Learning	0.00556	0.161	0.00483

9. Grad-CAM Visualization

Grad-CAM was applied to the **EfficientNet-B2 classification model** to visualize important regions in X-ray images.

- Heatmaps highlight areas influencing predictions
- Red regions indicate high importance

These visualizations help in understanding model decisions and improve interpretability.

Example output:

- Pneumonia detection
- Lung opacity regions
- Normal cases

10. Error Analysis

Missed Abnormalities

Small and low-contrast abnormalities were sometimes not detected.

False Detections

The model occasionally detected abnormalities near ribs and diaphragm regions.

Localization Errors

Multiple overlapping bounding boxes occurred at low confidence thresholds.

Classification Confusion

Some diseases with similar visual patterns were misclassified.

11. Conclusion

In Milestone-3, the CliniScan system was successfully improved using:

- Hyperparameter tuning
- Advanced data augmentation
- Transfer learning
- Model evaluation
- Grad-CAM visualization

Although performance improvements were observed, challenges remain in detecting subtle abnormalities and improving localization accuracy.